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Course: Data Science
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SQL Aggregate Functions & Operators

THE STUDENTS TABLE

1. SELECT DISTINCT department
FROM students;

Students

department
IT
HR
Finance

2. SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department;

Students

department	avg_age
IT	20.5
HR	22
Finance	23

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3 SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;

students

department	student_count
IT	2
HR	2

4. SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;

students

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finence
5	Eve	22	HR

5. SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT','HR') AND age > 21;

students

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

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THE COURSES Table

6. SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) >5;

Courses

department	total_credits
IT	11

7. SELECT course_id, course_name, department, credits
FROM Courses
WHERE credits != 4;

Courses

Course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

courses

Course_id	Course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

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9 SELECT max(grade) AS max_grade,
Min(grade) AS min_grade,
AVG(grade) AS avg_grade
FROM enrollments;

enrollments

max_grade	min_grade	avg_grade
90	78	84.6

10 SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;

enrollments

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

THE SALARIES TABLE

11 SELECT department,
SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus
FROM Salaries
GROUP BY department;

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11

Salaries

department	total_salary	total_bonus
IT	122 000	10500
HR	109 000	7500
Finance	7000	6000

12.
```
SELECT department, AVG(salary) AS avg_salary
FROM Salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

Salaries

department	avg_salary
IT	61000
Finance	70 000

13
```
SELECT employee_id,
       name,
       salary,
       bonus,
       (salary + bonus) AS total_compensation
FROM Salaries
WHERE (salary + bonus) > 60000;
```

Salaries

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65 000
3	Spike	70000	6000	76 000
4	Tyke	62000	5500	67500

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14 PROJECTS TABLE

14 SELECT department,
SUM(budget) AS total_budget,
AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;

Projects

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15 SELECT project_id,
project_name,
department,
budget
FROM projects
WHERE budget BETWEEN 50000 AND 120 000
AND department != 'Marketing';

Projects

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000

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